

Data Mining and Knowledge Discovery

Databases today can range in size into the terabytes or petabytes. Within these masses of data lies hidden information of strategic importance. This explosive growth of available data volume is a result of the computerization of our society and the fast development of powerful computing, sensing and data collection, storage, and publication tools. But when there is such enormous amount of data how could we draw any meaningful conclusion? We are data rich, but information poor. The abundance of data, coupled with the need for powerful data analysis tools, has been described as a *data rich but information poor* situation.

Data mining is being used both to increase revenues and to reduce costs. The potential returns are enormous. Innovative organizations worldwide are using data mining to locate and appeal to higher-value customers, to reconfigure their product offerings to increase sales, and to minimize losses due to error or fraud.

What is data mining?

Mining is a brilliant term characterizing the process that finds a small set of precious bits from a great deal of raw material.

Definition:

- Data mining is the process of discovering meaningful new correlations, patterns and trends by sifting through large amounts of data stored in repositories, using pattern recognition technologies as well as statistical and mathematical techniques.
- Data mining is the analysis of (often large) observational data sets to find unsuspected relationships and to summarize the data in unique ways that are both understandable and useful to the data owner.
- Data mining is an interdisciplinary field bringing together techniques from machine learning, pattern recognition, statistics, databases, and visualization to address the issue of information extraction from large database.

Many other terms carry a similar or slightly different meaning to data mining, such as *knowledge mining from data*, *knowledge extraction*, *data/pattern analysis*, *data archaeology*, and *data dredging*.

Many people treat data mining as a synonym for another popularly used term, Knowledge Discovery from Data, or KDD. While others view data mining as simply an essential step in Knowledge discovery.

Applications:

Market Basket Analysis Fraud Detection Intrusion Detection

Customer Segmentation Bio Informatics

Knowledge Discovery in Database (KDD)

1. **Data Cleaning** - noise and inconsistent data is removed.
2. **Data Integration** - multiple data sources are combined.
3. **Data Selection** - data appropriate to the analysis task are retrieved from the database.
4. **Data Transformation** - data are transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.
5. **Data Mining** - intelligent methods are applied in order to extract data patterns.
6. **Pattern Evaluation** - data are evaluated to identify the truly interesting patterns.
7. **Knowledge Presentation** - knowledge is represented using visualization techniques.

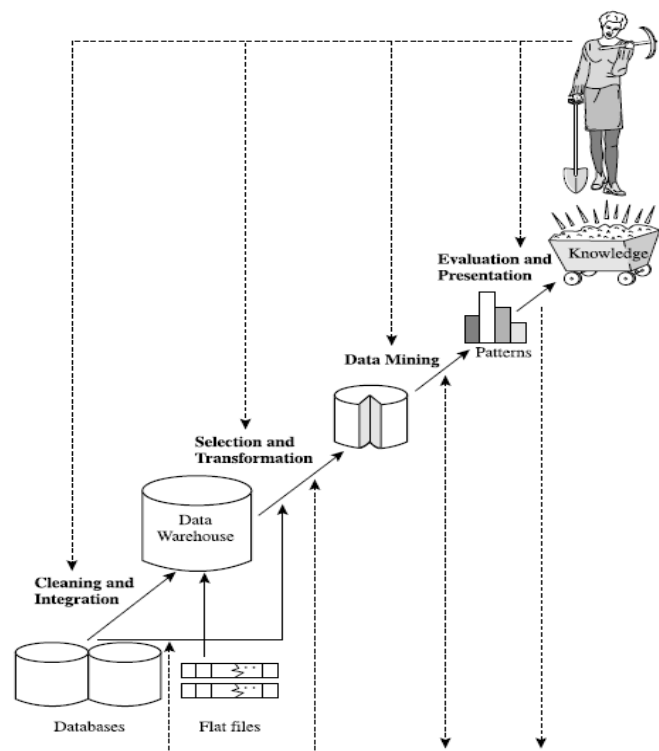


Figure. Data mining as a step in Knowledge Discovery

Steps 1 to 4 are different forms of data preprocessing, where the data are prepared for mining. The data mining step may interact with the user or a knowledge base. The interesting patterns are presented to the user and may be stored as new knowledge in the knowledge

base. So, according to this view, data mining is only one step in entire knowledge discovery process. However, in industry, in media, and in the database research area, the term data mining is becoming more popular than the longer term of knowledge discovery from data.

Based on this view, the architecture of a typical data mining system has following major components:

Data sources: Data Sources consists of one or a set of databases, data warehouses, spreadsheets, Word Wide Web or other kinds of information repositories. Data cleaning and data integration techniques may be performed on the data.

Database or data warehouse server: The database or data warehouse server is responsible for fetching the relevant data, based on the user's data mining request.

Knowledge base (KB): a knowledge base is a centralized repository for information: a public library or a database of related information about a particular subject. A KB is not a static collection of information, but a dynamic resource that may itself have the capacity to learn, as part of an artificial intelligence (AI) expert system.

Data mining engine: The data mining engine is the core component of any data mining system. It consists of a number of modules for performing data mining tasks including association, classification, characterization, clustering, prediction, time-series analysis etc.

Pattern evaluation module: The pattern evaluation module is mainly responsible for the measure of interestingness of the pattern by using a threshold value. It interacts with the data mining engine to focus the search towards interesting patterns.

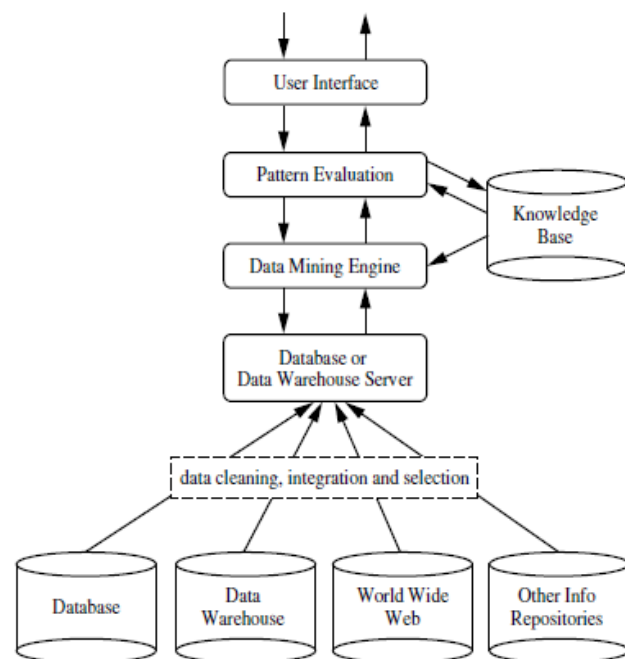


Figure. A Typical Data Mining Architecture

User interface: This module communicates between users and the data mining system, allowing the user to interact with the system by specifying a data mining query or task, providing information to help focus the search, and performing exploratory data mining based on the intermediate data mining results. In addition, this component allows the user to browse database and data warehouse schemas or data structures, evaluate mined patterns, and visualize the patterns in different forms.

Mining various kinds of knowledge

Different kinds of patterns and knowledge can be uncovered via data mining. In general, data mining tasks can be put into two categories: **descriptive data mining** and **predictive data mining**. Descriptive mining characterizes properties of the interested set of data, whereas predictive mining performs induction on the data set in order to make predictions.

Multidimensional data summarization

It is often tedious for a user to go over the details of a large set of data. Thus it is desirable to automatically summarize an interested set of data and compare it with the contrasting sets at some high levels. Such summaritive description of an *interested set of data* is called **data summarization**. Data summarization can often be conducted in a multidimensional space. If the multidimensional space is well defined and frequently used, such as product category, producer, location, or time, massive amounts of data can be aggregated in the form of **data cubes** to facilitate user's drill-down or roll-up of the summarization space with mouse clicking. The output of such multidimensional summarization can be presented in various forms, such as **pie charts**, **bar charts**, **curves**, **multidimensional data cubes**, and **multidimensional tables**, including crosstabs.

For structured data, multidimensional aggregation methods have been developed to facilitate such precomputation or online computation of multidimensional aggregations using data cube technology.

Mining frequent patterns, associations, and correlations

Frequent patterns, as the name suggests, are patterns that occur frequently in data. There are many kinds of frequent patterns, including frequent itemsets, frequent subsequences

(also known as sequential patterns), and frequent substructures. A *frequent itemset* typically refers to a set of items that often appear together in a transactional data set—for example, milk and bread, which are frequently bought together in grocery stores by many customers. A frequently occurring subsequence, such as the pattern that customers tend to purchase first a laptop, followed by a computer bag, and then other accessories, is a (*frequent*) *sequential pattern*. A substructure can refer to different structural forms (e.g., graphs, trees, or lattices) that may be combined with itemsets or subsequences. If a substructure occurs frequently, it is called a (*frequent*) *structured pattern*. Mining frequent patterns leads to the discovery of interesting associations and correlations within data.

Example: Association analysis. Suppose that, a webstore manager wants to know which items are frequently purchased together (i.e., in the same transaction). An example of such a rule, mined from the transactional database, is

$\text{buys}(X, \text{"computer"}) \Rightarrow \text{buys}(X, \text{"webcam"})$ [support = 1%, confidence = 50%],

where X is a variable representing a customer. A **confidence**, or certainty, of 50% means that if a customer buys a computer, there is a 50% chance that she will buy webcam as well. A 1% support means that 1% of all the transactions under analysis show that computer and webcam are purchased together.

Classification and regression for predictive analysis

Classification is the process of finding a **model** (or function) that describes and distinguishes data classes or concepts. The model is derived based on the analysis of a set of **training data** (i.e., data objects for which the class labels are known). The model is used to predict the class labels of objects for which the class labels are unknown.

Depending on the classification methods, a derived model can be in various forms, such as a set of *classification rules* (i.e., *IF-THEN rules*), a *decision tree*, a *mathematical formula*, or a learned *neural network*. A **decision tree** is a flowchart-like tree structure, where each node denotes a test on an attribute value, each branch represents an outcome of the test, and tree leaves represent classes or class distributions. Decision trees can easily be converted to classification rules. A **neural network**, when used for classification, is typically a collection of neuron-like processing units with weighted connections between the units. There are many

other methods for constructing classification models, such as naïve Bayesian classification, support vector machines, and k -nearest-neighbor classification.

Whereas classification predicts categorical (discrete, unordered) labels, **regression** models continuous-valued functions. That is, regression is used to predict missing or unavailable *numerical data values* rather than (discrete) class labels. The term *prediction* refers to both numeric prediction and class label prediction. **Regression analysis** is a statistical methodology that is most often used for numeric prediction, although other methods exist as well. Regression also encompasses the identification of distribution *trends* based on the available data.

Cluster analysis

Unlike classification and regression, which analyze class-labeled (training) data sets, **cluster analysis** (also called **clustering**) groups data objects without consulting class labels. In many cases, class-labeled data may simply not exist at the beginning. Clustering can be used to generate class labels for a group of data. The objects are clustered or grouped based on the principle of *maximizing the intraclass similarity and minimizing the interclass similarity*. That is, clusters of objects are formed so that objects within a cluster have high similarity in comparison to one another, but are rather dissimilar to objects in other clusters. Each cluster so formed can be viewed as a class of objects, from which rules can be derived.

Clustering can also facilitate **taxonomy formation**, that is, the organization of observations into a hierarchy of classes that group similar events together.

Outlier analysis

A data set may contain objects that do not comply with the general behavior or model of the data. These data objects are **outliers**. Many data mining methods discard outliers as noise or exceptions.

However, in some applications (e.g., fraud detection) the rare events can be more interesting than the more regularly occurring ones. The analysis of outlier data is referred to as **outlier analysis** or **anomaly mining**.

Outliers may be detected using statistical tests that assume a distribution or probability model for the data, or using distance measures where objects that are remote from any other

cluster are considered outliers. Rather than using statistical or distance measures, density-based methods may identify outliers in a local region, although they look normal from a global statistical distribution view.

Data Types and Attributes

A data is a known fact that can be recorded and have implicit (imbedded) meaning. For example, the names, eye color, telephone numbers, and addresses of the students of a class. So, data is a collection of data objects and their attributes.

A collection of attributes describe an object

Attribute

An attribute is a property or characteristic of an object. Attribute values are numbers or symbols assigned to an attribute. Different attributes can be mapped to the same set of values. Example: Attribute values for ID and age are integers but properties of attribute values can be different. ID has no limit but age has a maximum and minimum value.

Properties and types of attribute values:

The type of attribute depends on which of the following properties it possesses:

Distinctness: = and $\neq$

Order: $<$, $\leq$, $>$ and $\geq$

Addition: + and -

Multiplication: * and /

A simple way to specify the type of an attribute is to identify the properties of numbers that corresponds to those properties. Given these properties we can define following types of attributes:

i Categorical Attributes

Categorical attributes are also called qualitative attributes. Categorical attributes lack most of the properties of numbers even if they are represented by numbers. If these attributes are integers, they should be treated more like symbols.

a. Nominal Attributes:

In a nominal scale of measurement numbers or names are assigned to objects where the different values indicate different objects. Those attributes have no real meaning other than differentiating between objects. For example student roll numbers. Because the roll number that a student has provides no insight into the student's grade or IQ or anything like that it just simply differentiates between students. So roll number 23 and roll number 25 doesn't mean that the student who has 25 is better or smarter or anything like that it just means they're not the same student as number 23.

b. Ordinal Attribute

Ordinal attributes are the values which have meaningful order of ranking between them. For example the place someone finishes in a race first, second, third, and so on. If we know the place that they finished we know how they did relative to others. So, the first place person did better than second, second did better than third, and so on.

ii Numeric Attributes

Numeric attributes are represented by numbers and have most of the properties of numbers. Numeric attributes are also called quantitative attributes. These attributes can be either integer-valued or continuous.

a. Interval Scaled attribute

Interval scaled attribute are the values which are measured in a scale of equal size units between adjacent categories. For example temperature in Celsius scale. So if we take temperature in degrees Celsius the difference between 78 degrees and 79 degrees or that one degree difference is the same as the difference between 45 degrees and 46 degrees.

b. Ratio Scaled attributes

The ratio scaled attributes have meaningful intervals and also a true zero point. True zero point means the absence of the property like 0 inches means there is no length. For example weight in KG and here 10 KG is twice as much as five KG (i.e. $10 / 5$ is 2 or 2 times as much). So that's a meaningful ratio and in addition to this zero pounds means no weight or an absence of weight so there's a true zero point.

Attribute type		Description	Example	Properties
Categorical (Qualitative)	Nominal	The values of nominal attributes are just different symbol or name of things	zip codes, employee ID numbers, gender	Distinctness
	Ordinal	The values of ordinal attributes are possible values that have meaningful order or ranking among them	order of finishing a race, grades in exam	Distinctness and order
Numeric (Quantitative)	Interval	Interval scaled attributes are measured on a scale of equal size units	calendar dates, temperature in Celsius or Fahrenheit	Distinctness, order and Addition
	Ratio	The values of ratio scaled attributes has a true zero point	frequency of words in a document	Distinctness, order, addition and multiplication

Statistics of data

For data preprocessing to be successful, it is essential to have an overall picture of your data. Basic statistical descriptions can be used to identify properties of the data and highlight which data values should be treated as noise or outliers. There are three areas of basic statistical descriptions:

Measures of central tendency: Measure the location of the middle or center of a data distribution.

Dispersion of the data: How are the data spread out?

Graphic displays of basic statistics of data: use graphic displays of basic statistical descriptions to visually inspect our data.

Measuring the central tendency

There are various ways to measure central tendency of data. Suppose that we have some attribute X , like *salary*, which has been recorded for a set of objects. Let $x_1, x_2, \dots, x_N$ be the set of N observed values or *observations* for X . Here, these values may also be referred to as the data set (for X). If we were to plot the observations for *salary*, where would most of the values fall? This gives us an idea of the central tendency of the data. Measures of central tendency include the mean, median, mode, and midrange.

The most common and effective numeric measure of the “center” of a set of data is the (*arithmetic*) *mean*. Let $x_1, x_2, \dots, x_N$ be a set of N values or *observations*, such as for some numeric attribute X , like *salary*. The **mean** of this set of values is

$$\bar{x} = \frac{\sum_{i=1}^N x_i}{N} = \frac{x_1 + x_2 + \dots + x_N}{N}$$

Suppose we have the following values for *salary* (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12} = \frac{696}{12} = 58$$

Thus, mean salary is **\$58000**.

Sometimes, each value x_i in a set may be associated with a weight w_i for $i = 1, \dots, N$. The weights reflect the significance, importance, or occurrence frequency attached to their respective values. In this case, we can compute,

$$\bar{x} = \frac{\sum_{i=1}^N w_i x_i}{\sum_{i=1}^N w_i} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_N x_N}{w_1 + w_2 + \dots + w_N}$$

Although the mean is the single most useful quantity for describing a data set, it is not always the best way of measuring the center of the data. A major problem with the mean is its sensitivity to extreme (e.g., outlier) values. Even a small number of extreme values can

corrupt the mean. For example, the mean salary at a company may be substantially pushed up by that of a few highly paid managers. Similarly, the mean score of a class in an exam could be pulled down quite a bit by a few very low scores.

To offset the effect caused by a small number of extreme values, we can instead use the **trimmed mean**, which is the mean obtained after chopping off values at the high and low extremes. For example, we can sort the values observed for *salary* and remove the top and bottom 2% before computing the mean. We should avoid trimming too large a portion (such as 20%) at both ends, as this can result in the loss of valuable information.

For skewed (asymmetric) data, a better measure of the center of data is the **median**, which is the middle value in a set of ordered data values. It is the value that separates the higher half of a data set from the lower half.

In probability and statistics, the median generally applies to numeric data; however, we may extend the concept to ordinal data. Suppose that a given data set of N values for an attribute X is sorted in ascending order. If N is odd, then the median is the *middle value* of the ordered set. If N is even, then the median is not unique; it is the two middlemost values and any value in between. If X is a numeric attribute in this case, by convention, the median is taken as the average of the two middlemost values.

To compute median salary from the earlier data, it is first sorted in ascending order. Here, the data are already in ascending order. There is an even number of observations (i.e., 12); therefore, the median is not unique. It can be any value within the two middlemost values of 52 and 56 (that is, within the sixth and seventh values in the list). By convention, we assign the average of the two middlemost values as the median;

That is, $(52+56)/2 = 108/2 = 54$.

Thus, the median is \$54,000.

Suppose that we had only the first 11 values in the list. Given an odd number of values, the median is the middlemost value. This is the sixth value in this list, which has a value of \$52,000.

Mode is another measure of central tendency. The **mode** for a set of data is the value that occurs most frequently compared to all neighboring values in the set. Therefore, it can be determined for qualitative and quantitative attributes. It is possible for the greatest frequency to correspond to several different values, which results in more than one mode. Data sets with one, two, or three modes are respectively called **unimodal**, **bimodal**, and **trimodal**. In general, a data set with two or more modes is **multimodal**.

Example: The data from above example are bimodal. The two modes are \$52,000 and \$70,000.

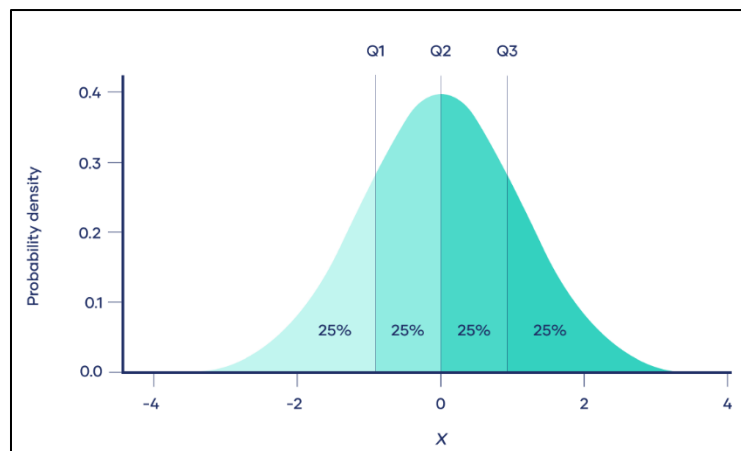
Measuring the dispersion of data

There are multiple measures to assess the dispersion or spread of numeric data. Some of them are range, quantiles, quartiles, percentiles, and the interquartile range. The five-number summary, which can be displayed as a boxplot, is useful in identifying outliers. Variance and standard deviation also indicate the spread of a data distribution.

Range, quartiles, and interquartile range

Let $x_1, x_2, \dots, x_N$ be the set of N observed values for some numeric attribute. The **range** of the set is the difference between the largest ($\max()$) and smallest ($\min()$) values.

Suppose that the data for attribute X are sorted in ascending numeric order. Imagine that we can pick certain data points so as to split the data distribution into equal-size consecutive sets, as in figure alongside. These data points are called *quantiles*. **Quantiles** are points taken at regular intervals of a data distribution, dividing it into essentially equal-size consecutive sets.



The quartiles give an indication of a distribution's center, spread, and shape. The **first quartile**, denoted by Q_1 , is the 25th percentile. It cuts off the lowest 25% of the data. The **third quartile**, denoted by Q_3 , is the 75th percentile—it cuts off the lowest 75% (or highest 25%) of the data. The second quartile is the 50th percentile. As the median, it gives the center of the data distribution. The distance between the first and third quartiles is a simple measure of spread that gives the range covered by the middle half of the data. This distance is called the **interquartile range (IQR)** and is defined as $IQR = Q_3 - Q_1$.

Five-number summary, boxplots, and outliers

No single numeric measure of spread (e.g., IQR) is very useful for describing skewed distributions. In the symmetric distribution, the median (and other measures of central tendency) splits the data into equal-size halves. This does not occur for skewed distributions. Therefore it is more informative to also provide the two quartiles Q_1 and Q_3 , along with the median. A common rule of thumb for identifying suspected **outliers** is to single out values falling at least $1.5 \times IQR$ above the third quartile or below the first quartile. Because Q_1 , the median, and Q_3 together contain no information about the endpoints (e.g., tails) of the data, a fuller summary of the shape of a distribution can be obtained by providing the lowest and highest data values as well. This is known as the *five-number summary*. The **five-number summary** of a distribution consists of the median (Q_2), the quartiles Q_1 and Q_3 , and the smallest and largest individual observations, written in the order of *Minimum, Q_1 , Median, Q_3 , Maximum*.

Example. Karen owns a bakery. She keeps track of how many items she sells each day. The number of items she sold over the last two weeks are: 25, 34, 28, 48, 52, 37, 29, 35, 54, 43, 64, 39, 58, 45. Summarize this data set with the five-number summary.

Step 1: Order the values from least to greatest.

Our list of values from least to greatest is: 25, 28, 29, 34, 35, 37, 39, 43, 45, 48, 52, 54, 58, 64

Step 2: Determine the minimum and maximum of the data set by identifying the lowest and highest values.

Based on our ordered list, the minimum value is **25** and the maximum value is **64**.

Step 3: Find the median of the data set. Separate the lower half from the upper half.

There are two middle numbers in our list of values: 39 and 43. To find the median we calculate $(39+43)/2=82/2=41$

Our median is **41**. We will use this to separate the data set into a lower half (25, 28, 29, 34, 35, 37, 39) and an upper half (43, 45, 48, 52, 54, 58, 64).

Step 4: Find the first and third quartiles by finding the median of the lower and upper half of the data.

The median, or middle number, of our lower half 25, 28, 29, 34, 35, 37, 39 is the value 34. Therefore our first quartile is **34**.

The median, or middle number, of our upper half 43, 45, 48, 52, 54, 58, 64 is the value 52. Therefore our third quartile is **52**.

Step 5: Summarize the data set by stating the minimum, first quartile, median, third quartile, and maximum.

Based on Steps 1 - 4, our five-number summary is:

- Minimum: 25
- First Quartile: 34
- Median: 41
- Third Quartile: 52
- Maximum: 64

Visualizing quartiles with boxplots

Boxplots are helpful visual summaries of a dataset. They're composed of boxes, which show the quartiles, and whiskers, which show the lowest and highest observations.

To make a boxplot, we first need to calculate the **five-number summary**:

Let n be the number of data values in the data set.

1. The minimum

- This is the lowest observation. If we order the numbers in your dataset from lowest to highest, the minimum is the first number. The minimum is sometimes called the zeroth quantile.

2. The first quartile or lower quartile

$$\text{Lower Quartile}(Q1) = \frac{1}{4}(n + 1)^{th} \text{term}$$

3. The second quartile or median

$$\text{Median}(Q2) = \frac{1}{2}(n + 1)^{th} \text{term}$$

4. The third quartile or upper quartile

$$\text{Upper Quartile}(Q3) = \frac{3}{4}(n + 1)^{th} \text{term}$$

5. The maximum

- This is the highest observation. If we order the numbers in your dataset from lowest to highest, the maximum is the last number. The maximum is sometimes called the fourth quantile.

With these five numbers, we can draw a boxplot:

Example.

Age(years)	1	2	3	4	5	6
frequency	2	3	4	1	2	2

First, we need to compute five number summary as:

Step 1: Count the number of observations in the dataset

$$n = 2 + 3 + 4 + 1 + 2 + 2 = 14$$

Step 2: Sort the observations in increasing order

1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 5, 5, 6, 6

Step 3: Find the first quartile

$$\text{Lower Quartile}(Q1) = \frac{1}{4}(14 + 1)^{th} \text{term} = (3.75)^{th} \text{term}$$

3.75 is not an integer, so Q1 is the number at position 4.

1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 5, 5, 6, 6

Q1 = 2 years

Step 4: Find the second quartile

$$\text{Median}(Q2) = \frac{1}{2}(14 + 1)^{th} \text{term} = (7.5)^{th} \text{term}$$

7.5 is not an integer, so Q2 is at position 8.

1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 5, 5, 6, 6

Q2 = 3 years

Step 5: Find the third quartile

$$\text{Upper Quartile}(Q3) = \frac{3}{4}(14 + 1)^{th} \text{term} = (11.25)^{th} \text{term}$$

11.25 is not an integer, so Q3 is the number at position 12.

1, 1, 2, 2, 2, 3, 3, 3, 3, 4, 5, 5, 6, 6

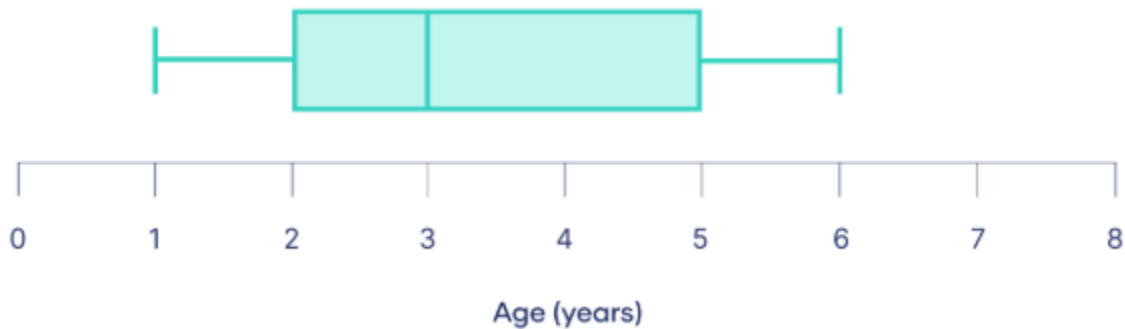
Q3 = 5 years

So, the five-number summary is:

1. The minimum: 1 year old
2. The first quartile: 2 years old

3. The second quartile: 3 years old
4. The third quartile: 5 years old
5. The maximum: 6 years old

These numbers determine the position of the box and whiskers on a number line:



Outliers

When dealing with a moderate number of observations, it is worthwhile to plot potential outliers individually. **Outlier** is a value that lies in a data series on its extremes, which is either very small or large and thus can affect the overall observation made from the data series. Outliers are also termed as **extremes** because they lie on the either end of a data series. Outliers are usually treated as **abnormal values** that can affect the overall observation due to its very high or low extreme values and hence should be discarded from the data series.

To do this in a boxplot, the whiskers are extended to the extreme low and high observations *only if* these values are less than $1.5 \times IQR$ beyond the quartiles. Otherwise, the whiskers terminate at the most extreme observations occurring within $1.5 \times IQR$ of the quartiles.

The **Interquartile range (IQR)** is the spread of the middle 50% of the data values.

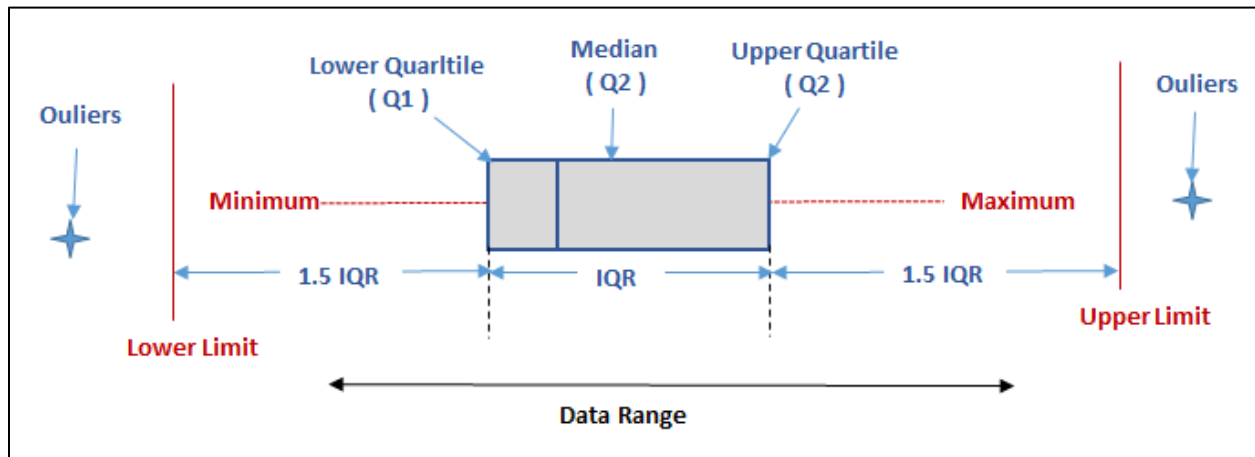
$$\text{Interquartile Range (IQR)} = \text{Upper Quartile (Q3)} - \text{Lower Quartile (Q1)}$$

$$IQR = Q3 - Q1$$

$$\text{Lower Limit} = Q1 - 1.5 IQR.$$

$$\text{Upper Limit} = Q3 + 1.5 IQR$$

The remaining cases are plotted individually. Boxplots can be used in the comparisons of several sets of compatible data.



Example

Let the data range be 130, 135, 199, 236, 236, 269, 271, 278, 278, 278, 283, 283, 291, 301, 301, 301, 303, 341, 450

Therefore $n = 19$

$$\text{Median}(Q2) = \frac{1}{2}(19 + 1)^{\text{th}} \text{ term} = 10^{\text{th}} \text{ term}$$

So, $Q2 = 278$

$$\text{Lower Quartile}(Q1) = \frac{1}{4}(19 + 1)^{\text{th}} \text{ term} = 5^{\text{th}} \text{ term}$$

So, $Q1 = 236$

$$\text{Upper Quartile}(Q3) = \frac{3}{4}(19 + 1)^{\text{th}} \text{ term} = 15^{\text{th}} \text{ term}$$

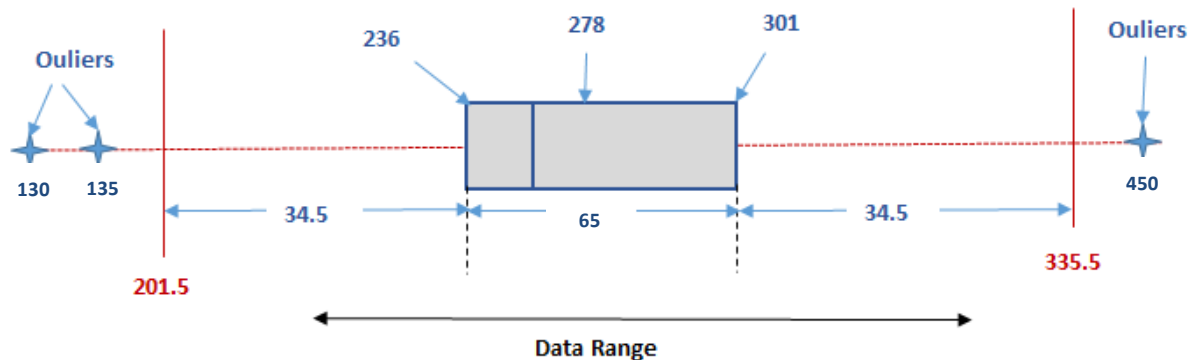
So, $Q3 = 301$

$$\text{Inter Quartile Range (IQR)} = Q3 - Q1 = 301 - 236 = 65$$

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR} = 236 - 1.5 \times 65 = 138.5$$

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR} = 301 + 1.5 \times 65 = 398.5$$

So, the box plot with outliers can be constructed as:



Variance and standard deviation

Variance and standard deviation are measures of data dispersion. They indicate how spread out a data distribution is. A low standard deviation means that the data observations tend to be very close to the mean, whereas a high standard deviation indicates that the data are spread out over a large range of values.

Variance and standard deviation both measure the variability of figures within a data set using the mean of a certain group of numbers. They are important to help determine volatility and the distribution of returns.

However, there are inherent differences between the two. While standard deviation measures the square root of the variance, the variance is the average of each point from the mean.

The **variance** of N observations, $x_1, x_2, \dots, x_N$ (when N is large), for a numeric attribute X is

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \left(\frac{1}{N} \sum_{i=1}^N x_i^2 \right) - (\bar{x})^2$$

Where, $\bar{x}$ is the mean value of the observations, The **standard deviation**, σ , of the observations is the square root of the variance, σ^2 .

Example. Suppose we have the following values for *salary* (in thousands of dollars), shown in ascending order: 30, 36, 47, 50, 52, 52, 56, 60, 63, 70, 70, 110

$$\bar{x} = \frac{30 + 36 + 47 + 50 + 52 + 52 + 56 + 60 + 63 + 70 + 70 + 110}{12} = \frac{696}{12} = 58$$

$$\text{So, } \sigma^2 = \frac{1}{12} (30^2 + 36^2 + 47^2 + \dots + 110^2) - 58^2 \approx 397.17$$

$$\sigma = \sqrt{397.17} \approx 19.47$$

Covariance and correlation analysis

In probability theory and statistics, correlation and covariance are two similar measures for assessing how much two attributes change together. Consider two numeric attributes A and B and a set of n realvalued observations $\{(a_1, b_1), \dots, (a_n, b_n)\}$. The mean values of A and B , respectively, are also known as the **expected values** on A and B , that is,

$$E(A) = \bar{A} = \frac{\sum_{i=1}^n a_i}{n}$$

And

$$E(B) = \bar{B} = \frac{\sum_{i=1}^n b_i}{n}$$

The covariance between A and B is defined as

$$\text{Cov}(A, B) = E((A - \bar{A})(B - \bar{B})) = \frac{\sum_{i=1}^n (a_i - \bar{A})(b_i - \bar{B})}{n}$$

Mathematically, it can be shown that

$$\text{Cov}(A, B) = E(A \cdot B) - \bar{A}\bar{B}$$

For two attributes A and B that tend to change together, if a value a_i of A is larger than $\bar{A}$ (the expected value of A), then the corresponding value of b_i of attribute B is likely to be larger than $\bar{B}$ (the expected value of B). Therefore the covariance between A and B is *positive*. On the other hand, if one of the attributes tends to be above its expected value when the other attribute is below its expected value, then the covariance of A and B is *negative*.

Example: Consider following table which presents a simplified example of stock prices observed at five time points for *ABCElectronics* and *XYZTech*, a high-tech company. If the stocks are affected by the same industry trends, will their prices rise or fall together?

	ABCElectronics(A)	XYZTech(B)
2013	1692	68
2014	1978	102
2015	1887	110
2016	2151	112
2017	2519	154

$$\text{Mean}(\text{ABCElectronics}) = \bar{A} = \frac{1692 + 1978 + 1887 + 2151 + 2519}{5} = 2045.4$$

$$\text{Mean}(\text{XYZTech}) = \bar{B} = \frac{68 + 102 + 110 + 112 + 154}{5} = 109.20$$

	ABCElectronics(A)	XYZTech(B)	$(a_i - \bar{A})$	$(b_i - \bar{B})$	$(a_i - \bar{A})(b_i - \bar{B})$
2013	1692	68	-353.4	-41.20	14554.88
2014	1978	102	-67.4	-7.20	485.28
2015	1887	110	-158.4	0.80	-126.72
2016	2151	112	105.6	2.80	295.68
2017	2519	154	473.6	44.80	21217.68
	$\bar{A} = 2045.4$	$\bar{B} = 109.20$			Total=36426.4

$$\text{Cov}(A, B) = \frac{36426.4}{5} = 7285.28$$

In such case, the positive covariance indicates that the price of the stock of ABCElectronics and the XYZTech tend to move in the same direction.

Correlation coefficient for numeric data

For numeric attributes, we can evaluate the correlation between two attributes, *A* and *B*, by computing the **correlation coefficient** (also known as **Pearson's product moment coefficient**,

named after its inventor, Karl Pearson). It is a statistical measure of the strength of a linear relationship between two variables. Its values can range from -1 to 1. A correlation coefficient of -1 describes a perfect negative, or inverse, correlation, with values in one series rising as those in the other decline, and vice versa. A coefficient of 1 shows a perfect positive correlation, or a direct relationship. A correlation coefficient of 0 means there is no linear relationship.

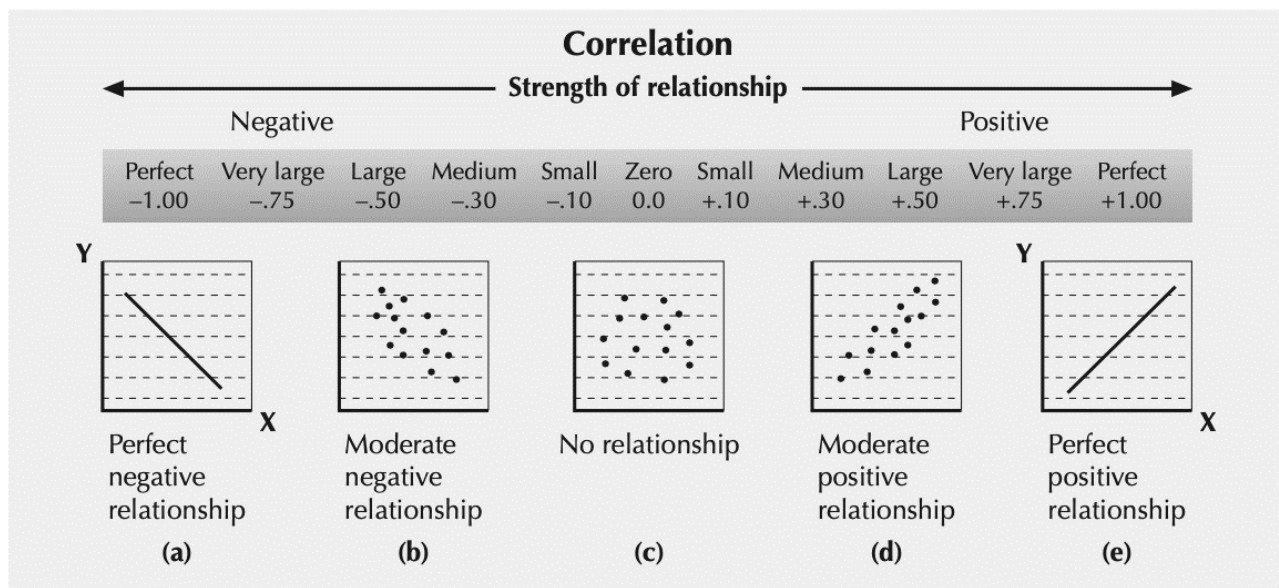
Correlation coefficients are used in science and finance to assess the degree of association between two variables, factors, or data sets. For example, as high oil prices are favorable for crude producers, one might assume that the correlation between oil prices and forward returns on oil stocks is strongly positive. Calculating the correlation coefficient for these variables based on market data reveals a moderate and inconsistent correlation over lengthy periods.

The Pearson coefficient uses a mathematical statistics formula to measure how closely the data points combining the two variables (with the values of one data series plotted on the x-axis and the corresponding values of the other series on the y-axis) approximate the line of best fit. The line of best fit can be determined through regression analysis.

The further the coefficient is from zero, whether it is positive or negative, the better the fit and the greater the correlation. The values of -1 (for a negative correlation) and 1 (for a positive one) describe perfect fits in which all data points align in a straight line, indicating that the variables are perfectly correlated.

In other words, the relationship is so predictable that the value of one variable can be determined from the matched value of the other. The closer the correlation coefficient is to zero, the weaker the correlation, until at zero no linear relationship exists at all.

If a correlation is negative (a), increases in one measure are associated with decreases in the other. (As Y gets larger, X gets smaller.) In a positive correlation (e), increases in one measure are associated with increases in the other. (As Y gets larger, X gets larger.) The center-left graph (b “moderate negative relationship”) might result from comparing time spent playing computer games (Y) with grades (X): More time spent playing computer games is associated with lower grades. The center graph (c “no relationship”) would result from plotting a person’s shoe size (Y) and his or her IQ (X). The center-right graph (d “moderate positive relationship”) could be a plot of grades in high school (Y) and grades in college (X) for a group of students: Higher grades in high school are associated with higher grades in college.



If x & y are the two variables of discussion, then the correlation coefficient can be calculated using the formula

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

Example

x	y	xy	x ²	y ²
12	2	24	144	4

15	4	60	225	16
18	6	94	324	36
21	8	168	441	64
27	12	324	729	144
$\Sigma x = 93$	$\Sigma y = 32$	$\Sigma xy = 670$	$\Sigma x^2 = 1863$	$\Sigma y^2 = 264$

On substituting values in the formula

$$r = 1$$

A correlation coefficient (r) of 1 indicates a perfect positive correlation between two variables (x and y). This means that as one variable increases, the other variable tends to increase as well in the same rate.

χ^2 Correlation test for nominal data

A **chi-squared test** (symbolically represented as χ^2) is basically a data analysis on the basis of observations of a random set of variables. Usually, it is a comparison of two statistical data sets. This test was introduced by **Karl Pearson** in 1900 for categorical data analysis and distribution. So it was mentioned as **Pearson's chi-squared test**.

Suppose A has c distinct values, namely, $a_1, a_2, \dots, a_c$, and B has r distinct values, namely, $b_1, b_2, \dots, b_r$. The data tuples described by A and B can be shown as a contingency table, with the c values of A making up the columns and the r values of B making up the rows. Let $(A_i B_j)$ denote the joint event that attribute A takes on value a_i and attribute B takes on value b_j , that is, where $(A = a_i, B = b_j)$. Each and every possible $(A_i B_j)$ joint event has its own cell (or slot) in the table.

The χ^2 value (also known as the *Pearson χ^2 statistic*) is computed as

$$\chi^2 = \sum_{i=1}^c \sum_{j=1}^r \frac{(o_{ij} - e_{ij})^2}{e_{ij}}$$

where o_{ij} is the *observed frequency* (i.e., actual count) of the joint event $(A_i B_j)$ and e_{ij} is the *expected frequency* of $(A_i B_j)$, which can be computed as

$$e_{ij} = \frac{\text{number of } a_i \times \text{number of } b_j}{n}$$

Where, n is the number of data tuples.

Example. Correlation analysis of nominal attributes using χ^2 . Suppose that a group of 1500 people was surveyed. The gender of each person was noted. Each person was polled as to whether his or her preferred type of reading material was fiction or nonfiction. Thus, we have two attributes, *gender* and *preferred_reading*. The observed frequency (or count) of each possible joint event is summarized in the contingency table shown in following table, where the numbers in parentheses are the expected frequencies. The expected frequencies are calculated based on the data distribution for both attributes using e_{ij} .

	male	female	total
Fiction	250(90)	200(360)	450
non_fiction	50(210)	1000(840)	1050
total	300	1200	1500

we can verify the expected frequencies for each cell. For example, the expected frequency for the cell (*male, fiction*) is

$$e_{11} = \frac{\text{number of male} \times \text{number of fiction}}{n} = \frac{300 \times 450}{1500} = 90$$

Notice that in any row, the sum of the expected frequencies must equal the total observed frequency for that row, and the sum of the expected frequencies in any column must also equal the total observed frequency for that column.

Now,

$$\begin{aligned} \chi^2 &= \frac{(250 - 90)^2}{90} + \frac{(50 - 210)^2}{210} + \frac{(200 - 360)^2}{360} + \frac{(1000 - 840)^2}{840} \\ &= 284.44 + 121.90 + 71.11 + 30.48 = 507.93 \end{aligned}$$

For this 2×2 table, the degrees of freedom are $(2 - 1) \times (2 - 1) = 1$. For 1 degree of freedom, the χ^2 value needed to reject the hypothesis at the 0.001 significance level is 10.828 (taken from the table of upper percentage points of the χ^2 distribution, typically available from any textbook on statistics).

Since our computed value is above this, we can reject the hypothesis that gender and referred_reading are independent and conclude that the two attributes are (strongly) correlated for the given group of people.

Similarity and distance measures

In data mining applications, such as clustering, outlier analysis, and nearest-neighbor classification, we need ways to assess how alike or unlike objects are in comparison to one another. For example, a store may want to search for clusters of *customer* objects, resulting in groups of customers with similar characteristics (e.g., similar income, area of residence, and age). Such information can then be used for marketing. A **cluster** is a collection of data objects such that the objects within a cluster are *similar* to one another and *dissimilar* to the objects in other clusters. Outlier analysis also employs clustering based techniques to identify potential outliers as objects that are highly dissimilar to others. Knowledge of object similarities can also be used in nearest-neighbor classification schemes where a given object (e.g., a *patient*) is assigned a class label (relating to, say, a *diagnosis*) based on its similarity toward other objects in the model.

The similarity and dissimilarity measures, are also referred to as measures of *proximity*. Similarity and dissimilarity are related. A similarity measure for two objects, i and j , will typically return value 0 if the objects are completely unlike. The higher the similarity value, the greater the similarity between objects. (Typically, a value of 1 indicates complete similarity, that is, the objects are identical.) A dissimilarity measure works the opposite way. It returns a value of 0 if the objects are the same (and therefore, far from being dissimilar). The higher the dissimilarity value, the more dissimilar the two objects are.

Data matrix vs. dissimilarity matrix

Suppose that we have n objects (e.g., persons, items, or courses) described by p attributes (also called *measurements* or *features*, such as age, height, weight, or gender). The objects are $x_1 = (x_{11}, x_{12}, \dots, x_{1p})$, $x_2 = (x_{21}, x_{22}, \dots, x_{2p})$, and so on, where x_{ij} is the value for object x_i of the j th attribute. For brevity, we hereafter refer to object x_i as object i . The objects may be tuples in a relational database and are also referred to as *data samples* or *feature vectors*.

Main memory-based clustering and nearest-neighbor algorithms typically operate on either of the following two data structures:

Data matrix (or object-by-attribute structure): This structure stores the n data objects in the form of a relational table or an n -by- p matrix (n objects $\times$ p attributes):

$$\begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Dissimilarity matrix (or object-by-object structure): This structure stores a collection of proximities that are available for all pairs of n objects. It is often represented by an n -by- n table:

$$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ \vdots & \vdots & \ddots & \\ d(n,1) & d(n,2) & \cdots & 0 \end{bmatrix}$$

where $d(i, j)$ is the measured **dissimilarity** or “difference” between objects i and j . In general, $d(i, j)$ is a nonnegative number that is close to 0 when objects i and j are highly similar or “near” each other, and becomes larger the more they differ. Note that $d(i, i) = 0$; that is, the difference between an object and itself is 0.

Proximity measures for nominal attributes

A nominal attribute can take on two or more states. For example, *map_color* is a nominal attribute that may have five states: *red*, *yellow*, *green*, *pink*, and *blue*.

Let the number of states of a nominal attribute be M . The states can be denoted by letters, symbols, or a set of integers, such as $1, 2, \dots, M$. Such integers are used just for data handling and do not represent any specific ordering. The dissimilarity between two objects i and j can be computed based on the ratio of mismatches:

$$d(i, j) = \frac{p - m}{p}$$

where m is the number of *matches* (i.e., the number of attributes for which i and j are in the same state), and p is the total number of attributes describing the objects. Weights can be assigned to increase the effect of m or to assign greater weight to the matches in attributes having a larger number of states.

For example, consider following sample dataset. The dissimilarity matrix can be computed as:

Sample data		Form of matrix	Dissimilarity matrix
OID	Value	$\begin{bmatrix} 0 & & & \\ d(2,1) & 0 & & \\ d(3,1) & d(3,2) & 0 & \\ d(4,1) & d(4,2) & d(4,3) & 0 \end{bmatrix}$	$\begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 1 & 1 & 0 & \\ 0 & 1 & 1 & 0 \end{bmatrix}$
1	A		
2	B		
3	C		
4	A		

From this, we see that all objects are dissimilar except objects 1 and 4 (i.e., $d(4, 1) = 0$).

Dissimilarity of numeric data

Different distance measures are used for computing the dissimilarity of objects described by numeric attributes. These measures include the *Euclidean*, *Manhattan*, and *Minkowski distances*.

The most popular distance measure is **Euclidean distance** (i.e., straight line or “as the crow flies”). Let $i = (x_{i1}, x_{i2}, \dots, x_{ip})$ and $j = (y_{i1}, y_{i2}, \dots, y_{ip})$ be two objects described by p numeric attributes.

The Euclidean distance between objects i and j is defined as

$$d(i, j) = \sqrt{(y_{i1} - x_{i1})^2 + (y_{i2} - x_{i2})^2 + \dots + (y_{ip} - x_{ip})^2}$$

Another well-known measure is the **Manhattan (or city block) distance**, named so because it is the distance in blocks between any two points in a city (such as 2 blocks down and 3 blocks over for a total of 5 blocks). It is defined as

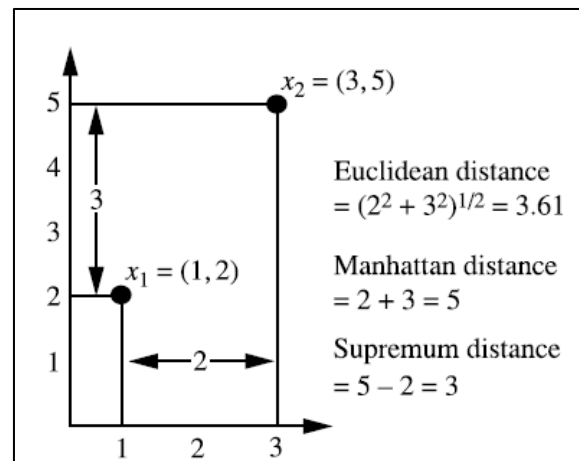
$$d(i, j) = |y_{i1} - x_{i1}| + |y_{i2} - x_{i2}| + \dots + |y_{ip} - x_{ip}|$$

Minkowski distance is a generalization of the Euclidean and Manhattan distances. It is defined as

$$d(i, j) = \sqrt[h]{|y_{i1} - x_{i1}|^h + |y_{i2} - x_{i2}|^h + \dots + |y_{ip} - x_{ip}|^h}$$

where h is a real number such that $h \geq 1$. (Such a distance is also called L_p **norm** in some literature, where the symbol p refers to our notation of h . We have kept p as the number of attributes) It represents the Manhattan distance when $h = 1$ (i.e., L_1 norm) and Euclidean distance when $h = 2$ (i.e., L_2 norm).

The **supremum distance**, also known as the **Chebyshev distance**, between two points is the maximum difference in their coordinates. For two points (x_1, y_1) and (x_2, y_2) in a 2-dimensional space, the supremum distance is calculated as: $\max(|x_2 - x_1|, |y_2 - y_1|)$



Cosine similarity

Cosine similarity measures the similarity between two vectors of an inner product space. It is measured by the cosine of the angle between two vectors and determines whether two vectors are pointing in roughly the same direction. It is often used to measure document similarity in text analysis.

Cosine similarity is a measure of similarity that can be used to compare documents or, say, give a ranking of documents with respect to a given vector of query words. Let $\mathbf{x}$ and $\mathbf{y}$ be two vectors for comparison. Using the cosine measure as a similarity function, we have

$$S_{cos}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Where $\|\mathbf{x}\|$ is the Euclidean norm of vector $\mathbf{x} = (x_1, x_2, \dots, x_p)$, defined as $\sqrt{(x_1)^2 + (x_2)^2 + \dots + (x_p)^2}$. Similarly, $\|\mathbf{y}\|$ is the Euclidean norm of vector $\mathbf{y}$.

For example,

Suppose $\mathbf{x} = (5, 0, 3, 0, 2, 0, 0, 2, 0, 0)$ and $\mathbf{y} = (3, 0, 2, 0, 1, 1, 0, 1, 0, 1)$. How similar are $\mathbf{x}$ and $\mathbf{y}$? On computing the cosine similarity between the two vectors, we get:

$$\begin{aligned} S_{cos}(\mathbf{x}, \mathbf{y}) &= \frac{(5 \times 3) + (0 \times 0) + (3 \times 2) + (0 \times 0) + (2 \times 1) + (0 \times 1) + (0 \times 0) + (2 \times 1) + (0 \times 0) + (0 \times 1)}{\sqrt{5^2 + 0^2 + 3^2 + 0^2 + 2^2 + 0^2 + 0^2 + 2^2 + 0^2 + 0^2} \sqrt{3^2 + 0^2 + 2^2 + 0^2 + 1^2 + 1^2 + 0^2 + 1^2 + 0^2 + 1^2}} \\ &= 0.94 \end{aligned}$$

Therefore if we were using the cosine similarity measure to compare these documents, they would be considered quite similar.

Data quality measures

Data have quality if they satisfy the requirements of the intended use. There are many factors comprising **data quality**, including *accuracy*, *completeness*, *consistency*, *timeliness*, *believability*, and *interpretability*.

Imagine that you are a manager at an online webstore and have been charged with analyzing the company's data with respect to your branch's sales. You immediately set out to perform this task. You carefully inspect the company's database and data warehouse, identifying and selecting the attributes or dimensions (e.g., *item*, *price*, and *units_sold*) to be included in your analysis. Alas! You notice that several of the attributes for various tuples have no recorded values. For your analysis, you would like to include information as to whether each item

purchased was advertised as on sale, yet you discover that this information has not been recorded. Furthermore, users of your database system have reported errors, unusual values, and inconsistencies in the data recorded for some transactions. In other words, the data you wish to analyze by data mining techniques are *incomplete* (lacking attribute values or certain attributes of interest, or containing only aggregate data); *inaccurate* or *noisy* (containing errors, or values that deviate from the expected); and *inconsistent* (e.g., containing discrepancies in the department codes used to categorize items).

This scenario illustrates three of the elements defining data quality: **accuracy**, **completeness**, and **consistency**. Inaccurate, incomplete, and inconsistent data are commonplace properties of large realworld databases and data warehouses. There are many possible reasons for inaccurate data (i.e., having incorrect attribute values). The data collection instruments used may be faulty. There may have been human or computer errors occurring at data entry. Users may purposely submit incorrect data values for mandatory fields when they do not wish to submit personal information (e.g., by choosing the default value “January 1” displayed for birthday). This is known as *disguised missing data*. Errors in data transmission can also occur. There may be technology limitations such as limited buffer size for coordinating synchronized data transfer and consumption. Incorrect data may also result from inconsistencies in naming conventions or data codes or inconsistent formats for input fields (e.g., *date*). Duplicate tuples also require data cleaning.

Incomplete data can occur for a number of reasons. Attributes of interest may not always be available, such as customer information for sales transaction data. Other data may not be included simply because they were not considered important at the time of entry. Relevant data may not be recorded due to a misunderstanding or because of equipment malfunctions. Data that were inconsistent with other recorded data may have been deleted. Furthermore, the recording of the data history or modifications may have been overlooked. Missing data, particularly for tuples with missing values for some attributes, may need to be inferred.

Recall that data quality depends on the intended use of the data. Two different users may have very different assessments of the quality of a given database. For example, a marketing analyst may need to access the database mentioned before for a list of customer addresses.

Some of the addresses are outdated or incorrect, yet overall, 80% of the addresses are accurate. The marketing analyst considers this to be a large customer database for target marketing purposes and is pleased with the database's accuracy, although as sales manager, you found the data inaccurate.

Timeliness also affects data quality. Suppose that you are overseeing the distribution of monthly sales bonuses to the top sales representatives in a company. Several sales representatives, however, fail to submit their sales records on time at the month-end. There are also a number of corrections and adjustments that flow in after the month-end. For a period of time following each month, the data stored in the database are incomplete. However, once all of the data are received, it is correct. The fact that the month-end data are not updated in a timely fashion has a negative impact on the data quality.

Two other factors affecting data quality are believability and interpretability. **Believability** reflects how much the data are trusted by users, whereas **interpretability** reflects how easily the data are understood. Suppose that a database, at one point, had several errors, all of which have since been corrected. The past errors, however, had caused many problems for sales department users, and so they no longer trust the data. The data also use many accounting codes, which the sales department does not know how to interpret. Even though the database is now accurate, complete, consistent, and timely, sales department users may regard it as of low quality due to poor believability and interpretability.

Data cleaning

Data cleaning routines work to “clean” the data by filling in missing values, smoothing noisy data, identifying or removing outliers, and resolving inconsistencies. If users believe the data are dirty, they are unlikely to trust the results of any data mining that has been applied to it. Furthermore, dirty data can cause confusion for the mining procedure, resulting in unreliable output. Although most mining routines have some procedures for dealing with incomplete or noisy data, they are not always strong. Therefore, a useful preprocessing step is to run the data through some data cleaning routines.

❖ Missing Data

Many tuples have no recorded value for several attributes. The values may be missing due to:

- Equipment malfunctioning
- Inconsistent with other data and hence deleted
- Not entered due to misunderstanding
- Certain data may be considered unimportant while entering.

Ways to handle missing data

- *Ignore the tuple*: This is usually done when the class label is. This method is not very effective, unless the tuple contains several attributes with missing values.
- *Fill in the missing value manually*: In general, this approach is time-consuming and may not be feasible given a large data set with many missing values.
- **Fill missing values automatically with:**
 - *Global constant*: Replace all missing attribute values by the same constant, such as a label like “Unknown”.
 - *Attribute mean*: For example, suppose that the average income of customers is Rs.16,000. Use this value to replace the missing value for income.
 - *Most probable value*: use inference-based techniques such as Bayesian formula or decision tree.

❖ Noisy data

Noise is a random error or variance in a measured variable. For a numerical attribute such as, price, we can “smooth” out the data to remove the noise by following techniques:

a. Binning

Binning methods smooth a sorted data value by consulting its “neighborhood,” that is, the values around it. The sorted values are distributed into a number of “buckets,” or bins. Because binning methods consult the neighborhood of values, they perform local smoothing.

For example,

Let us consider following data for price: 4, 15, 28, 34, 8, 21, 9, 21, 29, 26, 24, 25

First, Sort the data for price (in dollars) as: 4, 8, 9, 15, 21, 21, 24, 25, 26, 28, 29, 34

Partition into equal-frequency bins:

Here, number of Bins=3, so, each Bin has $12/3=4$ items.

Bin 1: 4, 8, 9, 15

Bin 2: 21, 21, 24, 25

Bin 3: 26, 28, 29, 34

i Smoothing by bin means:

Bin 1: 9, 9, 9, 9

Bin 2: 23, 23, 23, 23

Bin 3: 29, 29, 29, 29

ii Smoothing by bin boundaries:

Bin 1: 4, 4, 4, 15

Bin 2: 21, 21, 25, 25

Bin 3: 26, 26, 26, 34

b. Regression:

Data can be smoothed by fitting the data to a function, such as with regression. Linear regression involves finding the “best” line to fit two attributes (or variables), so that one attribute can be used to predict the other. Multiple linear regression is an extension of linear regression, where more than two attributes are involved and the data are fit to a multidimensional surface.

c. Clustering:

Outliers may be detected by clustering, where similar values are organized into groups, or “clusters.” Intuitively, values that fall outside of the set of clusters may be considered outliers.

Data Integration

Data integration combines data from multiple sources into a coherent data store, as in data warehousing. These sources may include multiple databases, data cubes, or flat files.

There are a number of issues to consider during data integration.

- *Entity identification problem:* The entity identification problem arises when different sources have different values for same attributes. For example, customer id in one database and cust_number in another may refer to the same attribute.
- *Data redundancy:* Data is said to be redundant if same data appears in various locations. Inconsistencies in attribute or dimension naming can cause redundancies in the resulting data set.

The data redundancy problem can be handled by correlation analysis. Given two attributes, such analysis can measure how strongly one attribute implies the other, based on the available data.

Data transformation

In data transformation, the data are transformed or combined into forms appropriate for mining. Data transformation can involve the following:

- i **Smoothing:** which works to remove noise from the data. Such techniques include binning, regression, and clustering.
- ii **Aggregation:** summary or aggregation operations are applied to the data. For example, the daily sales data may be aggregated so as to compute monthly and annual total amounts.
- iii **Generalization:** low-level or “primitive” (raw) data are replaced by higher-level concepts through the use of concept hierarchies. For example, values for numerical attributes, like age, may be mapped to higher-level concepts, like youth, middle-aged, and senior.

iv **Normalization:** the attribute data are scaled so as to fall within a small specified range, such as -1.0 to 1.0 or 0.0 to 1.0.

a. **Min-max normalization:**

Suppose that $\min_A$ and $\max_A$ are the minimum and maximum values of an attribute, A. A Min-max normalization maps a value, v , of A to v' in the range $[\text{new_min}_A, \text{new_max}_A]$ by computing

$$v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

Eg. Let the income range is 12000 to 98000 be normalized to $[0.0, 1.0]$. Then 73600 is normalized to:

$$\frac{73600 - 12000}{98000 - 12000} (1.0 - 0) + 0 = 0.716$$

b. **z-score normalization:**

In z-score normalization (or zero-mean normalization), the values for an attribute, A, are normalized based on the mean and standard deviation of A. A value, v , of A is normalized to v' by computing

$$v' = \frac{v - \text{mean}_A}{\sigma_A}$$

Eg. Let $\text{mean} = 54000$ and $\text{Standard Deviation } (\sigma) = 16000$. Then 73600 is mapped to

$$\frac{73600 - 54000}{16000} = 1.225$$

c. **Normalization by decimal scaling:**

It normalizes the value by moving the decimal point of values of attribute A. The number of decimal points moved depends on the maximum absolute value of A. A value, v , of A is normalized to v' by computing

$$v' = \frac{v}{10^j}$$

Where j is the smallest integer such that $\text{Max}(|v'|) < 1$.

Eg. Suppose that the recorded values of A ranges from -986 to 917. The maximum absolute value of A is 986. To normalize by decimal scaling, we therefore divide each value by 1,000 (i.e., $j = 3$) so that -986 normalizes to -0.986 and 917 normalizes to 0.917.

Data reduction

Data reduction techniques can be applied to obtain a reduced representation of the data set that is much smaller in volume, yet closely maintains the integrity of the original data. That is, mining on the reduced data set should be more efficient yet produce the same (or almost the same) analytical results.

Strategies for data cube reduction:

a. Data cube aggregation:

Aggregation operation is performed to combine two or more objects into a single object. For example the data consist of the sales per quarter, for the years 2002 to 2004. If the interest of user is in the annual sales (total per year), rather than the total per quarter. The data can be aggregated so that the resulting data summarize the total sales per year instead of per quarter. The resulting data set is smaller in volume, without loss of information necessary for the analysis task.

2002		2003		2004	
Quarter	Sales	Quarter	Sales	Quarter	Sales
Q1	50,000	Q1	45,000	Q1	50,000
Q2	25,000	Q2	40,000	Q2	45,000
Q3	15,000	Q3	15,000	Q3	40,000
Q4	20,000	Q4	25,000	Q4	35,000



year	Sales
2002	110,000
2003	125,000
2004	170,000

b. Dimensionality reduction:

In dimensionality reduction, data encoding or transformations are applied so as to obtain a reduced or “compressed” representation of the original data.

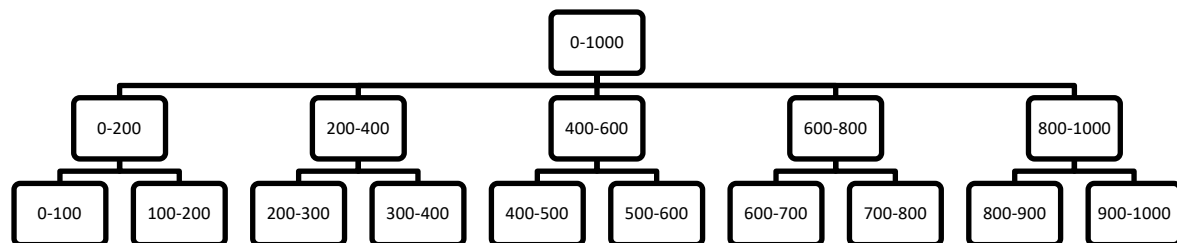
c. Data compression:

Generally done for space optimization on mostly multimedia data.

1. Data discretization

Raw data values for attributes are replaced by ranges or higher conceptual levels. Data discretization is a form of numerosity reduction that is very useful for the automatic generation of concept hierarchies.

Discretization and concept hierarchy generation are powerful tools for data mining, in that they allow the m

**Origin of Data Mining**

The term "Data mining" was introduced in the 1990s, but data mining is the evolution of a field with a long history. The traditional data analysis techniques have encountered following practical difficulties in meeting the challenges that motivated the development of data mining:

Scalability: because of advances in data generation and collection, data sets with sizes of gigabytes, terabytes or even petabytes are becoming common. If the data mining algorithms are to handle these massive data sets, then they must be scalable. For example scalability can be improved by using sampling or developing parallel and distributed algorithms.

High dimensionality: it is now common to encounter data sets with hundreds of attributes instead of the handful common a few decades ago. Data sets with temporal or spatial components often have high dimensionality. The traditional data analysis techniques that were developed for low-dimensional data often do not work well for such high dimensional data.

Heterogeneous and complex data: as the role of data mining in business, science, medicine and other field has grown, so has the need for technique that can handle heterogeneous attributes. The traditional data analysis techniques only deals with data sets containing attributes of the same type, either continuous or categorical.

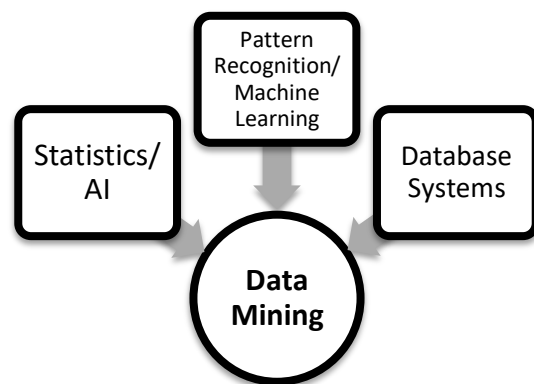
Data ownership and distribution: sometimes the data needed for an analysis is not stored in one location or owned by one organization. Instead, the data is geographically distributed among resources belonging to multiple entities. This requires the development of distributed data mining techniques.

Non-traditional analysis: the traditional statistical approach is based on a hypothesize-and-test paradigm. In this approach, a hypothesis is purposed, an experiment is designed to gather data and then the data is analyzed with respect to hypothesis. Unfortunately, this process is extremely labor intensive. Current data mining techniques require this task to be done in huge scale. So, the data mining techniques have been motivated by the desire to automate the process of hypothesis generation and evaluation.

So, to meet all those challenges, researchers from different disciplines began to focus on developing more efficient and scalable tools that could handle diverse types of data. This work, which culminated in the field of data mining, built upon the methodology and algorithms that researchers had previously used.

Data mining draws upon the ideas such as

- Sampling, estimation and hypothesis testing from statistics
- Search algorithm, modeling techniques and learning theories from artificial intelligence
- Various other ideas from pattern recognition and machine learning
- And database systems are of course needed for providing support for efficient storage, indexing and query processing.



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